

The Architecture of Astrometrics: Transitioning from Point-and-Click GUIs to Collaborative Spatial Computing

For decades, the dominant paradigm of human-computer interaction (HCI) has been dictated by the constraints of two-dimensional screens and peripheral pointing devices. The modern computing environment is heavily reliant on the Graphical User Interface (GUI), a system that has evolved into a monolithic architecture—effectively a pixel-perfect video feed pointed at a screen, operating atop tens of millions of lines of complex rendering code. This architecture creates persistent bottlenecks in both computational rendering and human cognitive processing, leading advanced users to actively seek alternatives.

In popular science fiction, particularly within the *Star Trek* universe, the complete absence of the computer mouse is a notable design choice. When characters in *Star Trek: Voyager* engage in high-stakes navigation—such as plotting courses through the uncharted Delta Quadrant using imperfect, dynamically updating, and critically flawed sensor data—they do not rely on desktop GUIs. Instead, they utilize the Astrometrics lab (or Stellar Cartography), an immersive, room-sized volumetric environment originally designed and integrated by the Borg character, Seven of Nine¹. The necessity of this "big room" was not merely aesthetic or theatrical; it was portrayed as a fundamental requirement for collaborative sensemaking. Analyzing spatial relationships, avoiding anomalies at warp speed, and making critical topological decisions based on simulated data required shared spatial-analytic states where multiple domain experts could perceive, manipulate, and interpret information simultaneously, without the isolating encumbrance of head-mounted displays¹.

This report investigates the immediate trajectory of user interface design, bridging the gap between highly efficient, keyboard-driven inputs (Terminal User Interfaces) and collaborative, spatial outputs (Immersive Analytics, CAVE systems, and Light Field Displays). By examining the cognitive friction of GUIs, the computational overhead of modern web rendering, and the emergence of volumetric photon-based displays, this analysis delineates the UI of the future that is actively being constructed today—a framework that closely mirrors the conceptual foundations of the Astrometrics lab.

The Bottleneck of the Pointer: GUIs, TUIs, and Cognitive Friction

To understand the UI of the future, it is first necessary to diagnose the mechanical and cognitive failures of the current GUI paradigm. While GUIs are highly intuitive for novices because they rely on visual recognition rather than memory recall, they impose strict upper limits on the operational efficiency of expert users⁵. The fundamental limitation of the mouse and the GUI is formalized by two foundational models in human-computer interaction: Fitts's

Law and the Keystroke-Level Model (KLM).

Fitts's Law and the Inefficiency of Aimed Movement

Formulated by Paul Fitts in 1954, Fitts's Law quantifies the time required for a human to rapidly move to a target area, stating that the time to acquire a target grows with its distance and shrinks with its width⁶. The mathematical expression is generally represented as

$T = a + b \log_2(D/S + 1)$, where D is the distance to the target, S is the target size, and the logarithmic term represents the "index of difficulty" measured in bits⁷.

Fitts's Law models aimed movement as a closed-loop motor control system⁷. When an individual uses a mouse to point at a screen, the human motor system must issue an initial command, evaluate the remaining distance visually, and make continuous micro-adjustments to correct the error margin. This biological feedback loop inherently throttles execution speed. In a dense GUI environment, users are constantly subjected to this penalty. As interfaces become more visually complex, information density often collapses, distances between interactive elements grow, and targets become smaller, exponentially increasing the index of difficulty⁶. Navigating a cascading menu or seeking a specific button requires visual search, the application of the steering law (moving through a confined graphical tunnel), and continuous motor correction⁷. This results in cognitive friction that repeatedly interrupts the user's operational flow state.

The Keystroke-Level Model and the Empirical Cost of the Mouse

The inefficiency of pointer-based interaction is further quantified by the Keystroke-Level Model (KLM), developed in the 1980s by Card, Moran, and Newell⁶. The KLM calculates task execution time by decomposing an interface interaction into a sequence of pre-established, primitive physical and mental operators¹⁰.

The model assigns empirical time values to specific user actions based on extensive psychological testing⁹. When evaluating a standard GUI action—such as moving the hand from the keyboard to the mouse, visually searching for a button, pointing to the target, and finally clicking—the cumulative latency is immense compared to executing a memorized keyboard shortcut.

Keystroke-Level Model (KLM) Operator Latencies

OPERATOR	DESCRIPTION	EST. TIME (S)
K	Keying: The time it takes to tap a key on the keyboard or a button on the mouse (skilled typist).	0.20
P	Pointing: The time it takes to move the mouse to a position on the display.	1.10
H	Homing: The time it takes to move the hand from the keyboard to the mouse or from the mouse to the keyboard.	0.40
M	Mental preparing: The time it takes to mentally prepare for the next step.	1.35

Mouse Homing & Pointing Cost (H + P): **1.50 s**

Standard empirical time estimates for basic physical and cognitive operators in the Keystroke-Level Model. The requirement to home the hand (H) and point the mouse (P) introduces significant latency compared to sequential keystrokes (K).

Data sources: [MIT 6.813 \(2018\)](#), [GOMS Task Analysis Overview](#), [MIT 6.813 \(2016\)](#), [DePaul University](#)

The data indicates that a standard keystroke operator (K) requires approximately 0.20 seconds, whereas pointing with a mouse (P) requires an average of 1.10 seconds—a direct consequence of Fitts's Law. Furthermore, the physical act of homing the hands between the keyboard and the mouse (H) requires 0.40 seconds, and the mental preparation (M) needed to locate a target on a screen averages 1.35 seconds⁹. Therefore, a simple action that forces a user to reach for a mouse, find a target, and click can introduce over 3.0 seconds of latency, whereas a multi-key terminal shortcut executes in under a second¹².

The TUI Resurgence: Returning to Keyboard-Driven Efficiency

This fundamental inefficiency explains the growing resurgence of Terminal User Interfaces (TUIs) among expert users, particularly software developers and system administrators¹³. While early command-line interfaces (CLIs) functioned merely as stateless transducers (accepting an input string and outputting a text result), modern TUIs combine the stateful, screen-oriented behavior of GUIs with the raw speed of terminal deployment¹⁵.

A TUI bypasses the heavy accessibility trees and Document Object Model (DOM) coordinate

systems of GUIs, rendering a two-dimensional array of character cells directly to a pseudo-terminal driven entirely by raw byte streams and keystrokes¹⁵. Applications like LazyVim and LazyGit leverage complex keybindings to compress elaborate actions into rapid keystroke (K) operators, entirely eliminating the homing (H) and pointing (P) penalties¹³. Furthermore, TUI applications exhibit massive computational efficiency. While a modern Electron-based GUI running basic applications might consume gigabytes of RAM simply to maintain its state and web rendering engines, a TUI equivalent operates on a fraction of those resources. Users report running multiple instances of advanced TUIs utilizing less than 1 GB of memory and negligible CPU overhead¹⁵. This allows users to rely on terminal multiplexers to bounce rapidly between open sessions, maximizing input bandwidth and avoiding the resource bloat that plagues modern web-based interfaces¹⁶.

The Computational Overhead of the Web DOM

The inefficiency of the GUI is not purely biological; it is heavily rooted in modern software architecture. The web browser, which currently serves as the de facto operating system for most consumer and enterprise applications, relies on the Document Object Model (DOM). Rendering a standard GUI via the DOM involves a deeply nested and synchronous pipeline: the browser must parse HTML and CSS, construct a render tree, calculate precise mathematical layouts (reflow), and finally paint the pixels to the screen¹⁷.

Every time an interactive element changes, the browser is forced to traverse this massive DOM tree. This triggers synchronous reflows and repaints. This architecture, accurately described as a "pixel-perfect video" generated by millions of lines of code, introduces severe latency overhead. This limitation becomes glaringly obvious when developers attempt to render complex, high-frequency data, dynamic simulations, or 3D visualizations within a standard browser window¹⁸. Native applications bypass this overhead by communicating directly with the Graphics Processing Unit (GPU) via low-level APIs like Metal on macOS or Vulkan on Linux, dramatically accelerating the rendering pipeline¹⁹.

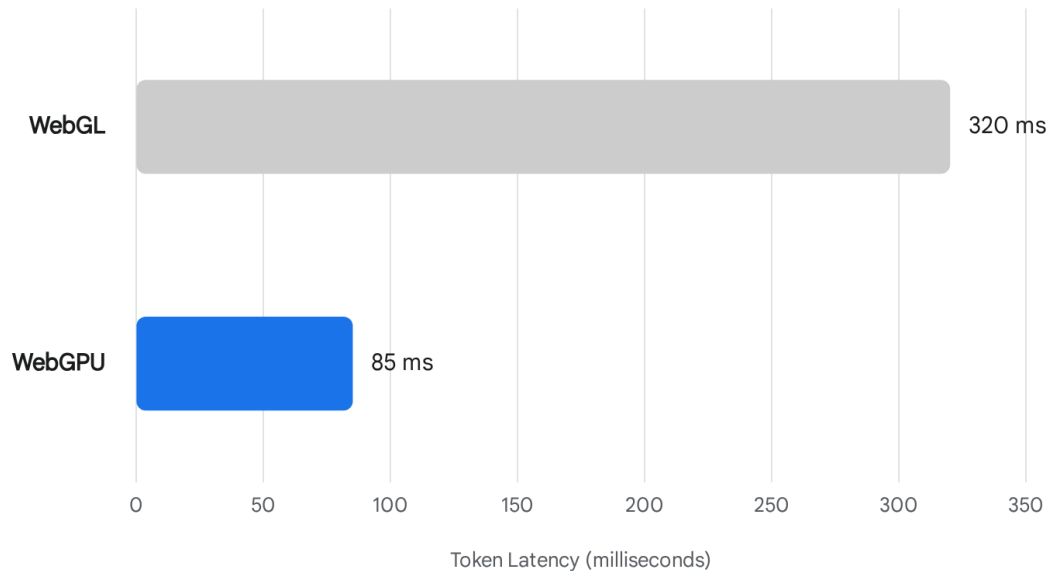
The Transition from WebGL to WebGPU

Recognizing that the DOM is insufficient for the future of UI—especially for spatial computing and high-dimensional analytics—the web architecture is currently undergoing a structural shift. WebGPU has emerged as a modern graphics API that provides low-overhead, native-like access to GPU hardware directly from the browser, succeeding the older WebGL standard²⁰. Benchmarks comparing WebGL to WebGPU reveal staggering performance disparities. WebGPU delivers 3x to 8x faster GEMM (General Matrix Multiply) execution for large matrices (e.g., 2048x2048 arrays)²¹. This acceleration is achieved because WebGPU provides access to native compute shaders, storage buffers, and workgroup shared memory—capabilities that WebGL fundamentally lacks²¹.

For intensive client-side workloads, such as running Large Language Model (LLM) inference directly in the browser, the impact is profound. Hardware benchmarking on Apple M2 architectures demonstrates that token latency drops from approximately 320 milliseconds

under WebGL down to 85 milliseconds when transitioning to WebGPU²¹.

Client-Side Inference Latency: WebGL vs. WebGPU



Performance benchmarking of token latency (milliseconds) during client-side Large Language Model inference (Phi-3-mini on Apple M2). WebGPU leverages native compute shaders to bypass traditional web rendering bottlenecks, reducing latency by over 73%.

Data source: [Sitepoint](#)

This computational paradigm shift is critical for the evolution of user interfaces. The UI of the future requires rendering massive volumetric datasets in real-time. If the render thread remains blocked by continuous DOM reflow calculations, spatial computing is impossible. By transitioning to WebGPU, the ecosystem is building the foundational pipeline required to manipulate "photons and lasers" at scale, paving the way for data visualization environments that reflect the complexity of the physical world²⁰.

The Astrometrics Paradigm: Navigating Imperfect, High-Dimensional Data

If TUIs and intent-driven keyboard operations represent the future of high-speed *input*, it is necessary to determine what represents the future of data *output*. As datasets grow exponentially in dimensionality and scale, representing them on 2D screens becomes fundamentally inadequate.

Consider the operational metaphor of Astrometrics in *Star Trek: Voyager*. The data the crew

relied upon was vast, constantly updating, and ultimately imperfect¹. A simulation of space required mapping stellar cartography, avoiding navigational hazards, and integrating fragmented sensor data across massive distances²³. Attempting to parse this data on a flat, 2D monitor using a scrolling viewport would invite catastrophic failure; the user would constantly lose the global context while zooming in on local anomalies. The "big room" was necessary because the scale and dimensionality of the data required spatial immersion to properly align human cognition with the complexity of the simulation¹.

The Limits of 2D and the Utility of 3D Visualization

The debate regarding the efficacy of 2D versus 3D visualization is a cornerstone of HCI research. Historically, information visualization researchers have been highly skeptical of 3D representations for abstract data²⁵. When early 3D idioms—such as cone trees, cam trees, or perspective walls—were forced onto flat 2D desktop screens in the 1990s, they frequently hindered user performance. Analysts struggled with issues like occlusion (where front elements hide rear elements), foreshortening, and a complete lack of true depth disparity²⁵.

Empirical studies have repeatedly demonstrated that for simple tasks requiring metric identification or single-variable lookups, 2D visualizations (like standard bar charts) consistently outperform 3D representations²⁶. For example, tests comparing 2D and 3D displays for diagnosing arterial disease showed that 2D environments led to significantly faster and more accurate responses when isolating specific metrics²⁷.

However, abstract tabular data is fundamentally different from spatial, topological, or volumetric data. High-dimensional scientific data—such as atmospheric models, spatial transcriptomics, LiDAR scans, or the dynamic network topologies of storm surges—has no natural scale or single optimal perspective³. When visualizing these structures, depth perception ceases to be a decorative element and becomes a critical, load-bearing channel of information²⁵.

The Emergence of Immersive Analytics (IA)

This necessity has given rise to the field of Immersive Analytics (IA). IA is formally defined as the science of analytical reasoning facilitated by immersive human-computer interfaces³¹. The discipline investigates how new interaction and display technologies—ranging from extended reality (XR) to CAVE systems and Light Fields—can support complex sensemaking⁴.

IA systems utilize immersion to generate environments where data can be visualized volumetrically, preserving the 3D nature of the data and reducing the cognitive effort required to mentally manipulate 2D slices into a cohesive 3D construct³³. By combining advanced data preprocessing (such as utilizing Python frameworks like Dask or Polars) with dimensionality reduction embeddings (like t-SNE or UMAP algorithms), IA allows analysts to physically navigate through dense point clouds of high-dimensional vectors³.

Spatial Memory and Cognitive Load in Immersive

Spaces

The effectiveness of the Astrometrics paradigm rests on human biology. The human brain did not evolve to parse scrolling lists on a glowing rectangle; it evolved to navigate, map, and recall spatial relationships within a physical environment. Effectively conducting visual analytical tasks relies heavily on the human capacity for visuo-spatial working memory—the ability to locate and recall the spatial arrangements of objects³⁵.

Leveraging Proprioception and Bodily Navigation

Immersive Analytics capitalizes on this biological hardware. By arraying information spherically or volumetrically around the user, IA systems offload cognitive processing directly into the physical environment. Psychological assessments, such as the Corsi block-tapping test, demonstrate that individuals exhibit enhanced spatial memory when they can utilize physical body activities to navigate data spaces³⁴.

Research assessing different 3D user interaction techniques indicates that physical locomotion—simply walking around a data set—provides superior spatial memory support compared to virtual "teleportation" or grabbing and rotating objects with controllers³⁴.

Proprioception (the body's ability to sense its location, movements, and actions) anchors the data in the user's mind. When data is placed in the viewer's physical environment, it allows domain experts to walk through data structures, inspect dense clusters from multiple angles, and utilize six degrees of freedom (6DoF) for natural manipulation³.

The Danger of Isolation in Head-Mounted Displays

However, placing users inside standard Virtual Reality Head-Mounted Displays (HMDs) introduces a critical flaw that undermines the purpose of the Astrometrics lab: isolation. When dealing with mission-critical data, decision-making is inherently collaborative. Domain experts must organize around shared spatial-analytic states that they can jointly perceive and interpret³.

Standard VR headsets isolate users from their physical surroundings and from the subtle non-verbal cues (pointing, gaze direction, body language) of their colleagues. While networked avatars and presence indicators attempt to solve this, they introduce friction in collaborative sensemaking³. The UI of the future requires a space where experts can look at the data, and simultaneously look at each other.

Building the Big Room: CAVEs and Hybrid Reality Environments

The technological realization of the Astrometrics lab exists today as the Cave Automatic Virtual Environment (CAVE). First developed at the Electronic Visualization Laboratory at the University of Illinois in 1992, the CAVE is a room-sized, multi-person, projection-based Virtual Reality theater³⁸.

A standard CAVE consists of a cube (typically 10x10x9 feet) where high-resolution stereoscopic graphics are rear-projected onto three to six walls, and often the floor³⁹. Unlike an HMD where the display is strapped to the face, the CAVE employs an "inside-out" viewing paradigm. The environment uses window projection, specifying the center of projection relative to the plane for each eye to create off-axis perspective projection³⁹.

The primary advantages of the CAVE architecture mirror the exact requirements of collaborative spatial computing:

- **Unencumbered Collaboration:** Because the massive displays are fixed at a distance from the viewer, multiple people can inhabit the space simultaneously⁴⁰. While one primary user's head position is actively tracked via infrared cameras to render the correct stereoscopic perspective dynamically, other users in the room can view the data in a shared physical context, behaving like "passengers in a bus" exploring the dataset together³⁹.
- **High Visual Acuity and Peripheral Vision:** Modern implementations, such as the VisCube systems, utilize direct-view LED panels or ultra-high-resolution digital projectors (e.g., 6000x4096 per wall) to provide immense pixel density and a panoramic field of view exceeding 180 degrees³⁹. This minimizes the loss of peripheral vision, which is a common and disorienting drawback of HMDs⁴².
- **Hybrid Reality Integration:** Advanced CAVE systems serve as Hybrid Reality Environments (HRE). They allow researchers to seamlessly transition between examining high-resolution 2D data elements (such as spreadsheets, code scripts, or static charts) and interacting with fully immersive 3D volumetric data within the same physical space⁴³.

Case Study: Submerge and Situated Flood Analytics

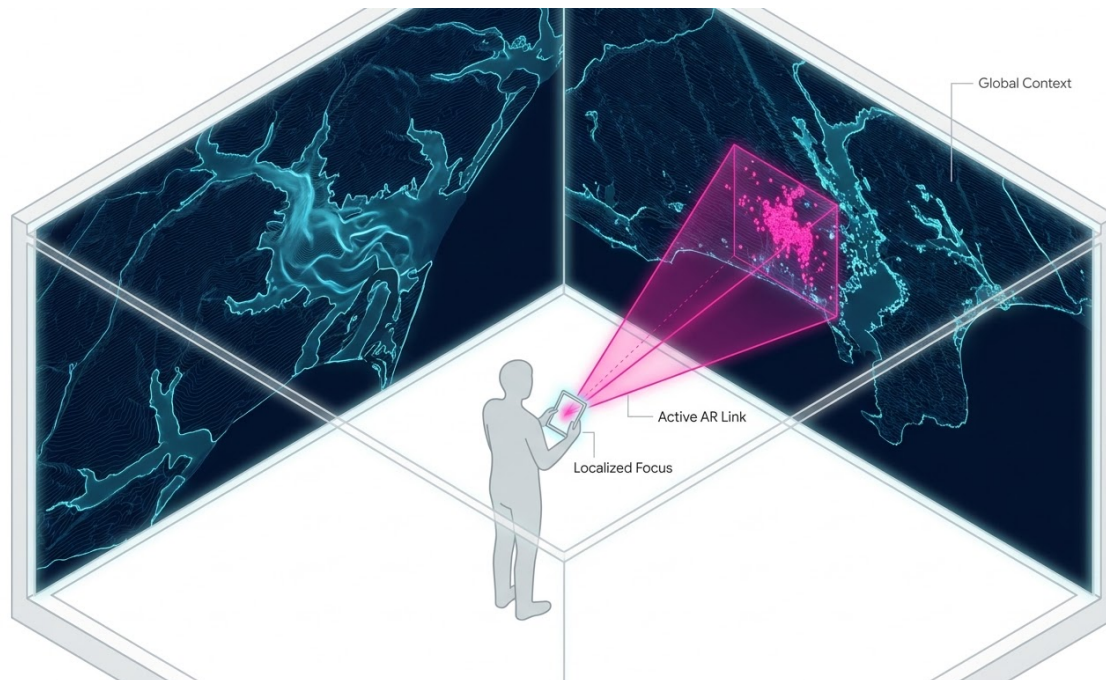
The application of CAVE environments is actively revolutionizing scientific analytics. A prominent example is the *Submerge* framework, developed through collaboration between computer scientists and atmospheric scientists to visualize 3D coastal and inland storm surge flooding scenarios²⁸. The system is deployed on massive immersive display ecologies, such as the gigapixel Reality Deck at Stony Brook University⁴⁴.

Historically, predicting flooding based on numerical simulations relied on 2D maps. While a 2D map conveys the bird's-eye extent of the water, it forces the user to infer depth, flow velocity, and volume purely from color gradients. In life-or-death emergency management scenarios, this cognitive leap can lead to severe misinterpretations of flood severity²⁸.

By rendering an adaptive height-field grid of the flood simulation inside a CAVE, *Submerge* surrounds emergency managers with a 1:1 scale representation of the water's depth and its interaction with 3D urban infrastructure⁴⁴. To solve the problem of interacting with data distributed across massive walls, the system employs a novel "Focus + Context" methodology. Analysts hold a video see-through Augmented Reality (AR) tablet that serves as an active proxy⁴⁴.

The AR tablet allows the user to perform localized visual analytics—manipulating parameters or requesting specific data queries on a focused point of interest—while the massive CAVE

screens maintain the global context of the entire storm system⁴⁴. This synthesis of mobile touch-interaction and room-scale display represents a significant leap forward in UI design, separating the mechanics of localized input from the cognitive necessity of global spatial awareness.



Photons and Light Fields: The Hardware of the Future

While CAVE systems offer unparalleled immersion and replicate the Astrometrics paradigm perfectly, they are bound by severe constraints: they require dedicated, expensive real estate, specialized non-magnetic construction materials in some cases, and complex multi-projector calibration arrays⁴¹. To bring volumetric UI out of specialized university basements and into standard corporate, medical, and scientific workspaces, the hardware must condense. The inquiry regarding whether the future UI is comprised of "photons and lasers in a room" is highly prescient. Technologically, this describes Light Field displays, which utilize highly engineered arrays of photons to recreate reality volumetrically, without the need for specialized 3D glasses, lasers, or Head-Mounted Displays.

The Looking Glass 65" Holographic Display

The most advanced commercial realization of this technology currently available is the Looking Glass 65" 8K holographic display⁴⁶. Representing a monumental shift in media consumption and UI design, the Looking Glass 65" is a massive, incredibly thin (3-inch) slab of glass, aluminum, and advanced optics capable of displaying deeply 3D holographic content to groups of up to 50 simultaneous viewers⁴⁷.

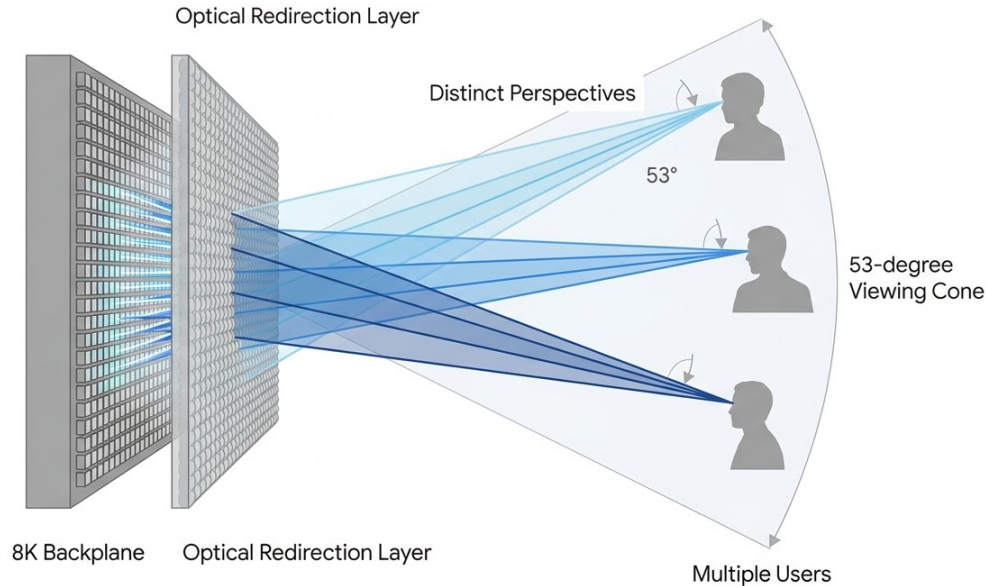
Unlike stereoscopic 3D televisions that require active shutter glasses, or HMDs that render two distinct images for a single tracked user, Light Field displays generate a true "holographic volume." The Looking Glass accomplishes this through a hybrid architecture combining an

ultra-high-resolution 8K (7680x4320) LCD or OLED backplane with a proprietary optical layer (lenticular lenses or specialized etching) that precisely redirects subpixels outward⁴⁸.

Light Field Display Specification	Technical Capability	Implication for UI Design
Input Resolution	8K (7680 x 4320 pixels)	Allows rendering of massive datasets with high visual acuity without pixelation ⁴⁸ .
Simultaneous Perspectives	45 to 100 discrete views	Enables true parallax; users can look "around" objects naturally by moving their heads ⁴⁸ .
Frame Rate	60 frames per second	Supports real-time dynamic simulations and fluid interaction models ⁴⁸ .
Viewing Cone	53 degrees	Accommodates large groups for collaborative sensemaking without tracking hardware ⁴⁸ .

Recreating Reality with Photons

By directing 100 million points of light every 60th of a second outward into a 53-degree viewing cone, the display synthesizes a physical light field⁴⁷. When multiple users stand in front of the display, their left and right eyes naturally intersect different angular projections of light. This natively generates binocular disparity and deep parallax without any tracking hardware⁴⁹. If a user moves their head to the left, they seamlessly transition into a new set of physical perspectives, looking "around" the 3D object exactly as they would in physical reality.



This hardware achieves the ultimate goal of the UI of the future: it provides the volumetric, spatial representation necessary for high-dimensional sensemaking while completely removing the physical barriers between collaborators⁴⁷. The system is highly compatible with modern rendering pipelines, integrating directly with 3D creation engines like Unity and Unreal via customized plugins⁴⁷. Furthermore, it relies heavily on the aforementioned advancements in GPU rendering. Calculating and pushing the massive array of views in real-time requires powerful hardware, such as the RTX 3090, to drive dual DisplayPort 8K signals effortlessly⁴⁸.

Software Infrastructure for Spatial Computing

Hardware displays, however advanced, require a standardized software layer to function effectively across diverse environments. As the ecosystem moves beyond the 2D web DOM, new protocols are being established.

OpenXR and WebXR serve as the foundation for this cross-platform interoperability. OpenXR continues to evolve beyond basic hardware abstraction, introducing features like Spatial Entities (anchors, planes, markers, and persistence)⁵⁴. These frameworks enable applications to understand and interact with a shared spatial context across sessions and devices⁵⁴. By utilizing WebXR libraries, developers can build holographic experiences directly on the web, linking spatial entity mapping with the low-latency rendering power of WebGPU⁴⁸. This standardization ensures that the volumetric UIs of the future are not siloed into proprietary applications, but form a cohesive spatial web.

Conclusion: Synthesizing the Next Interface

The inquiry regarding the step forward with the next UI is fundamentally a question of unbundling the input from the output. For forty years, the GUI forced a biological and

computational compromise: users were forced to utilize a highly latent 2D pointer to interact with a 2D screen, bottlenecked by the cognitive costs of Fitts's Law and the rendering overhead of the DOM.

The data unequivocally indicates that this paradigm is fracturing.

- **For Input and Execution:** Power users are retreating to the command line and Terminal User Interfaces (TUIs). Because human motor execution is constrained by physical pointing algorithms, the fastest way to convey intent to a machine is through memorized, semantic keybindings (as proven by the Keystroke-Level Model) rather than spatial pointing⁶. As models evolve, this intent-driven input will increasingly incorporate conversational voice interfaces and targeted AR proxies, further eliminating the need for a desktop mouse³¹.
- **For Output and Perception:** The necessity to visualize abstract, dynamic, and imperfect spatial data is driving the adoption of Immersive Analytics¹.

We are not building a world of floating, untouchable holograms manipulated by exhaustive, tiring hand gestures (a phenomenon known in HCI as "gorilla arm"). Instead, the UI of the future we can build *now* is a highly specialized, hybridized ecosystem.

Analysts will utilize highly efficient, tactile, and low-latency mechanical inputs (keyboards, TUIs, or localized AR tablets) to interface with the computational engine. The engine, leveraging APIs like WebGPU, will render massive datasets natively without the latency of the DOM¹⁹. The output will be directed into room-scale display ecologies—either massive multi-projector CAVE environments or 65-inch Light Field panels utilizing optical layers to redirect 100 million photons³⁹.

This architecture allows a team of researchers, commanders, or analysts to stand in a room, entirely unencumbered by headsets, viewing a true volumetric simulation of reality. They will converse naturally, point at physical manifestations of data, and make critical decisions based on shared visuo-spatial awareness. We are, piece by piece, assembling the Astrometrics lab.

Works cited

1. Star Trek Voyager (VOY) Season 7 Reviews - Ex Astris Scientia, <https://www.ex-astris-scientia.org/episodes/voy7.htm>
2. Starship Operations in a Star Trek MMORPG - Theory by Flatfingers, <https://flatfingers-theory.blogspot.com/2007/04/starship-operations-in-star-trek-mmorpg.html>
3. Exploring Collaborative Immersive Visualization & Analytics for High, <https://arxiv.org/html/2602.02743v2>
4. (PDF) Immersive Analytics 2.0: Spatial and Embodied Sensemaking, https://www.researchgate.net/publication/360408523_Immersive_Analytics_20_Spatial_and_Embodied_Sensemaking
5. what is the difference between gui and tui? - YouTube, <https://www.youtube.com/watch?v=BWUBYOf8pM>
6. Fitts's Law: Big and Near Beats Small and Far - UX Tigers, <https://www.uxtigers.com/post/fitts-law>

7. Reading 10: More Efficiency - MIT,
<https://web.mit.edu/6.813/www/sp18/classes/10-more-efficiency/>
8. Predictive Human Performance Modeling Made Easy,
<https://www.cs.drexel.edu/~dds26/publications/John-CHI04.pdf>
9. Relating Theory to Actual Results in Computer Science and Human,
<http://facweb.cs.depaul.edu/cmiller/hciEdDraft.pdf>
10. GOMS and KLM Task Analysis Overview | PDF - Scribd,
<https://www.scribd.com/document/695659319/GOMES>
11. (PDF) The Keystroke-Level Model for User Performance Time with,
https://www.researchgate.net/publication/220426122_The_Keystroke-Level_Model_for_User_Performance_Time_with_Interactive_Systems
12. Reading 4: Efficiency - MIT,
<https://web.mit.edu/6.813/www/sp16/classes/04-efficiency/>
13. TUI over GUI, and we might be onto something unprecedented,
<https://buttondown.com/fromthefringes/archive/tui-over-gui-and-we-might-be-onto-something/>
14. From CLI to GUI to TUI: Why developers are going back to Terminal,
<https://medium.com/@chrysophilist/from-cli-to-gui-to-tui-why-developers-are-going-back-to-terminal-c6a27aab1375>
15. Can LLMs Test Terminal User Interfaces? - arXiv,
<https://arxiv.org/html/2608.03743v1>
16. TUI vs GUI : r/opencode - Reddit,
https://www.reddit.com/r/opencode/comments/1w1vyk8/tui_vs_gui/
17. High-Performance and Energy-Efficient Mobile Web Browsing on,
https://edge.seas.harvard.edu/sites/g/files/omnuum6351/files/zhu10hpca_0.pdf
18. Performance comparison between webgl and webgpu #4448 - GitHub,
<https://github.com/gpuweb/gpuweb/discussions/4448>
19. Zed Editor: The Fastest AI-Powered Editor (2026),
<https://petronellatech.com/blog/zed-editor-setup-guide-2026/>
20. WebGPU: The Next Generation of Browser Graphics and Compute,
<https://blog.4dpipeline.com/webgpu-the-next-generation-of-browser-graphics-and-compute>
21. WebGPU vs. WebGL: Performance Benchmarks for Client-Side,
<https://daily.dev/posts/webgpu-vs-webgl-performance-benchmarks-for-client-side-inference-bxfaw5zzt>
22. ai-agency/examples/nexus-spatial-discovery.md at claude ... - GitHub,
<https://github.com/brandononchain/ai-agency/blob/claude/fork-ai-agency-vHHTT/ai-agency/examples/nexus-spatial-discovery.md>
23. Voyager brought back "treasure trove" of technologies , can ... - Reddit,
https://www.reddit.com/r/startrek/comments/1pq62li/voyager_brought_back_treasure_trove_of/
24. MUH051069 Star Trek Adventures - Delta Quadrant (Printer Friendly),
<https://www.scribd.com/document/454466547/MUH051069-Star-Trek-Adventures-s-Delta-Quadrant-Printer-Friendly-OEF-2020-pdf>
25. 2. Immersive Analytics: Time to Reconsider the Value of 3D for,

- https://vvisse.iat.sfu.ca/user/data/papers/IAchapter2_3DVis.pdf
26. Full article: 3D bivariate visualizations in immersive virtual reality (IVR),
<https://www.tandfonline.com/doi/full/10.1080/15230406.2026.2652392>
 27. Turning implicit into explicit: does it matter if the rainbow color,
<https://www.tandfonline.com/doi/full/10.1080/15230406.2025.2467316>
 28. Submerge: Visualizing Storm Surge Flooding Simulations in,
<https://www.computer.org/csdl/journal/tg/2024/09/10319326/1S5SPzLmWqI>
 29. Embodied Navigation in Immersive Abstract Data Visualization,
<https://www.computer.org/csdl/journal/tg/2021/02/09222098/1nTrQ1hHyyA>
 30. (PDF) Immersive analytics: An overview - ResearchGate,
https://www.researchgate.net/publication/363275717_Immersive_analytics_An_overview
 31. Immersive Analytics: Theory and Research Agenda - PMC,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7805807/>
 32. Immersive Analytics: Theory and Research Agenda - Frontiers,
<https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2019.00082/full>
 33. Immersive Visualisation and Exploration of Gesture Data,
https://angli.phd/assets/pdf/GestureExplorer_FullPaper.pdf
 34. arXiv:2308.13955v1 [cs.HC] 26 Aug 2023, <https://arxiv.org/pdf/2308.13955>
 35. Evaluating the Effectiveness of Spatial Memory in 2D and 3D,
https://www.researchgate.net/publication/2398144_Evaluating_the_Effectiveness_of_Spatial_Memory_in_2D_and_3D_Physical_and_Virtual_Environments
 36. (PDF) Effects of Display Layout on Spatial Memory for Immersive,
https://www.researchgate.net/publication/364290943_Effects_of_Display_Layout_on_Spatial_Memory_for_Immersive_Environments
 37. Evaluating 3D User Interaction Techniques on Spatial Working,
https://www.researchgate.net/publication/373450840_Evaluating_3D_User_Interaction_Techniques_on_Spatial_Working_Memory_for_3D_Scatter_Plot_Exploration_in_Immersive_Analytics
 38. Cave Automatic Virtual Environment - ResearchGate,
https://www.researchgate.net/publication/4045442_Cave_Automatic_Virtual_Environment
 39. THE CAVE" AUTOMATIC VIRTUAL ENVIRONMENT,
<https://ntrs.nasa.gov/api/citations/19960026482/downloads/19960026482.pdf>
 40. CAVE Systems - Visbox, Inc., <http://www.visbox.com/products/cave/>
 41. Cave automatic virtual environment - Wikipedia,
https://en.wikipedia.org/wiki/Cave_automatic_virtual_environment
 42. In-Depth Review of Augmented Reality: Tracking Technologies,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9824627/>
 43. Dataspace: A Reconfigurable Hybrid Reality Environment for ... - arXiv,
<https://arxiv.org/html/1903.03700v1>
 44. Submerge: Visualizing Storm Surge Flooding Simulations in,
https://saeedboorboor.github.io/papers/Submerge_TVCG_crc.pdf
 45. Demonstrating Immersive Analytical Tools with Augmented Reality,

- https://www.researchgate.net/publication/305486262_Situated_Analytics_Demonstrating_Immersive_Analytical_Tools_with_Augmented_Reality
46. Looking Glass Factory reveals 65-in 8K holographic display,
<https://www.hiddenwires.co.uk/news/article/looking-glass-factory-reveals-65in-8k-holographic-display>
 47. Looking Glass Factory Introduces the Largest Holographic Display,
<https://displaydaily.com/looking-glass-factory-introduces-the-largest-holographic-display-in-the-world-the-looking-glass-65/>
 48. Looking Glass Larger Display Sizes,
<https://lookingglassfactory.com/large-display-sizes>
 49. Looking Glass Factory launches 65-inch holographic display,
<https://gamesbeat.com/looking-glass-factory-launches-65-inch-holographic-display/>
 50. Looking Glass Factory's New Hololuminescent Display Means 3D,
<https://www.techeblog.com/looking-glass-factory-hololuminescent-display/>
 51. Looking Glass Holographic Light Field Displays: Experience Group,
<https://lookingglassfactory.com/displays-overview>
 52. #1106: Tribeca XR: Story Behind "Zanzibar" Volumetric Doc Shown,
<https://voicesofvr.com/1106-tribeca-xr-story-behind-zanzibar-volumetric-doc-shown-on-looking-glass-factorys-new-65-inch-holographic-display/>
 53. Looking Glass 65" Light Field Display,
<https://lfdocs.lookingglassfactory.com/getting-started/looking-glass-65>
 54. From Runtime Architecture to Spatial Computing by Frédéric Plourde,
<https://www.youtube.com/watch?v=HhZRIZ1JpOc>
 55. XR and Spatial Computing: Transforming the Future with OpenXR,
https://www.youtube.com/watch?v=iK0dKqCk_SA
 56. WebXR, A-Frame and Networked-Aframe as a Basis for an Open,
<https://arxiv.org/html/2404.05317v5>